

12.3 Classwork: Complex Numbers: Polar Form and de Moivre (no calculator)

Answer both problems in the spaces provided. Show your working. Give all answers as exact values (surds and fractions of π), not decimals.

1. Consider the complex numbers $z = -1 + i\sqrt{3}$ and $w = 2 - 2i$.

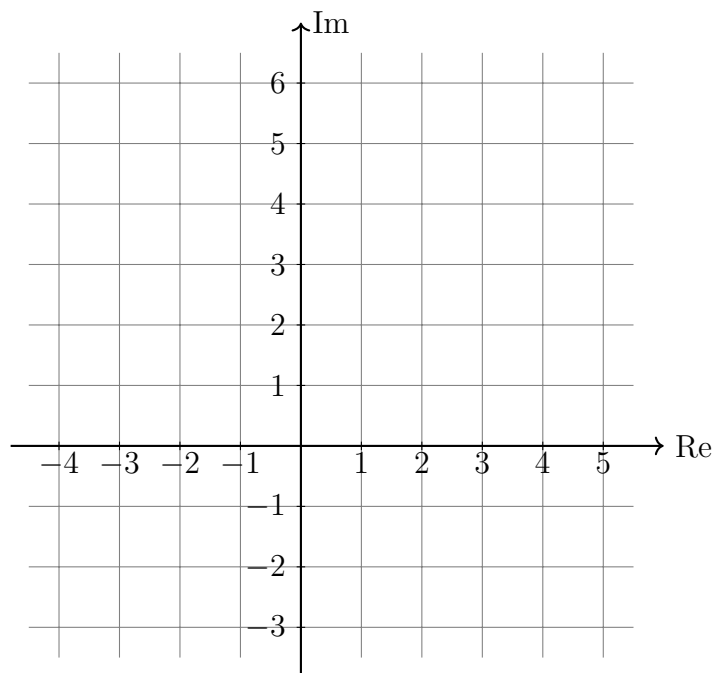
In parts (a) to (c), give every argument θ in the range $-\pi < \theta \leq \pi$.

(a) Write z in the form $r(\cos \theta + i \sin \theta)$ and in the form $re^{i\theta}$.

(b) Write w in the form $r(\cos \theta + i \sin \theta)$ and in the form $re^{i\theta}$.

(c) Hence find zw and $\frac{z}{w}$, giving each answer in the form $re^{i\theta}$.

(d) On the Argand diagram below, plot and label the points representing z , w and zw .



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2. (a) Use de Moivre's theorem to find the exact value of $(1 + i)^8$.
- (b) Find the three cube roots of $8i$, giving each answer in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.
- (c) State the geometric relationship between the three cube roots of $8i$ when they are plotted on an Argand diagram.

